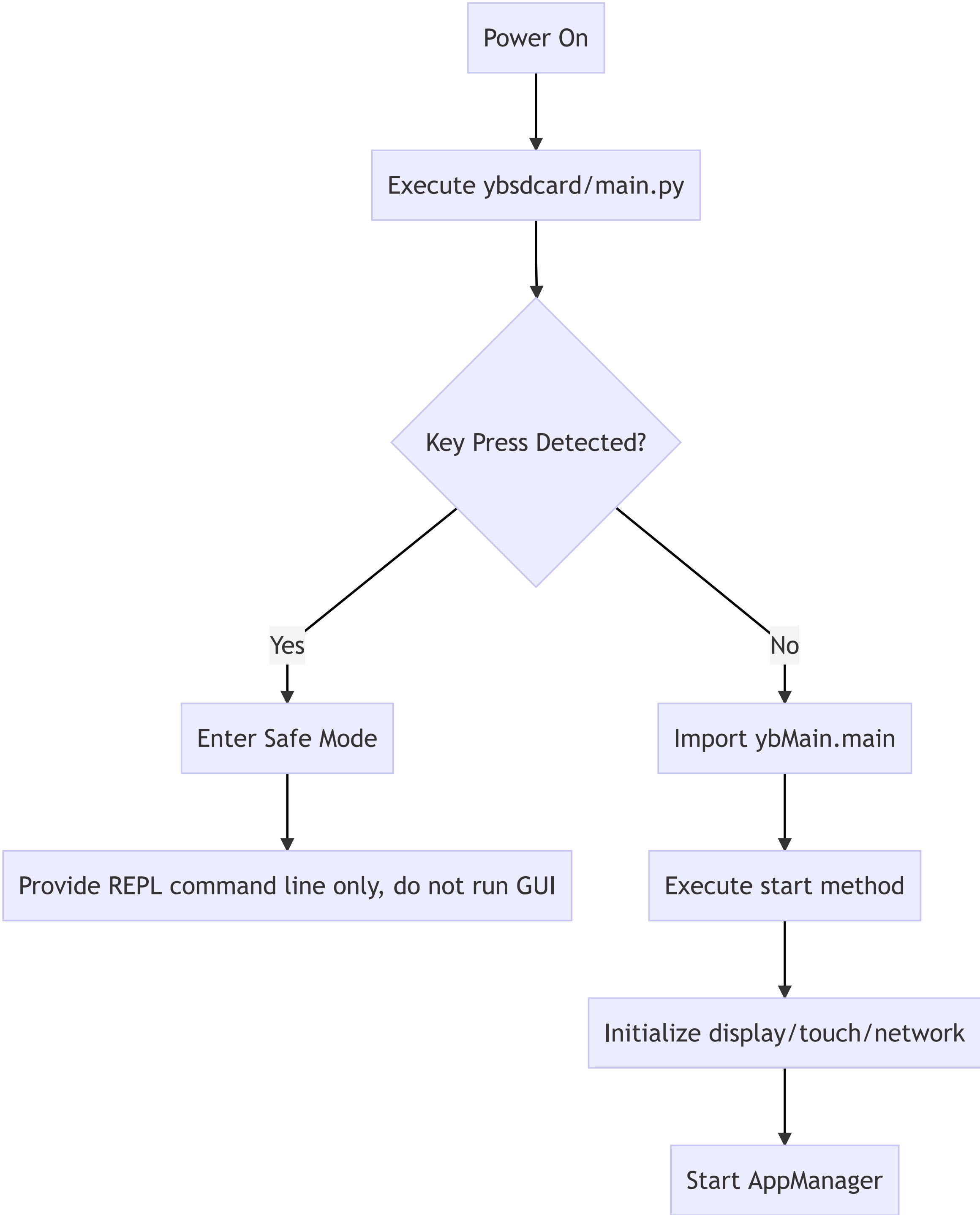
GUI System Architecture

1. Course Objectives

In this lesson, we will analyze the overall architecture of the Yahboom K230 GUI system

2. Architecture Diagram



3. Core Source Code Analysis

File Path: `src/ybsdcard/main.py`

This file is the **entry point** of the entire GUI system. When the device boots, the MicroPython firmware will automatically find and run `main.py`.

3.1 Safe Boot Mechanism (Anti-Brick Logic)

If you want to temporarily disable the GUI auto-start, or if there's a serious bug in `main.py` that causes the device to crash or freeze on boot, making it impossible to connect to the IDE for repair, you can skip the auto-start by pressing and holding the Key button for 3 seconds before powering on (press before power on --> power on --> release after 3 seconds).

K230's GUI system is designed with a safety backdoor**:

```
import time
from ybUtils.YbKey import YbKey

# Initialize the button
key = YbKey()
abort_main = 0

# Check button state
if key.is_pressed():
    time.sleep_ms(10) # Debounce
    if key.is_pressed():
        abort_main = 1 # If button is pressed, mark as abort startup
```

Code Analysis:

1. `YbKey()`: The system initializes a button object.
2. `key.is_pressed()`: Detects whether the user has pressed the function key at the moment of startup.
3. `abort_main = 1`: If a press is detected, the variable is set to 1, meaning "don't start GUI, return control to the user".

3.2 Starting the Core Engine

If the user has not pressed the function key, the system will continue to load the GUI core:

```
if not abort_main:
    from ybMain.main import *
    start()
```

Code Analysis:

1. `if not abort_main`: Only executes after the safety check passes.
2. `from ybMain.main import *`: Imports the core module `ybMain`. This is like when the Linux kernel finishes loading and starts `systemd` or the graphical interface.
3. `start()`: This is the startup function of the core engine. Once executed, the screen will light up and enter the graphical interface.